

Lab 2: Device's information

Pham Quang Duong - 2540048

October 1, 2026

The objective of this lab is to use `numba` to retrieve information about the available GPU, including the device name, device id, core information, and memory size.

The GPU information obtained is:

- Device ID: 0
- Device name: Tesla T4
- Compute Capability: 7.5
- Multiprocessor count: 40
- Total CUDA cores: $40 \times 64 = 2560$
- Memory Size: 14911.69 MB

The following figure shows the output obtained from the labwork.

```
Found 2 CUDA devices
id 0          Tesla T4          [SUPPORTED]
      Compute Capability: 7.5
      PCI Device ID: 4
      PCI Bus ID: 0
      UUID: GPU-7d59ea56-6b80-479b-7974-dbc2c59b5d4
      Watchdog: Disabled
      FP32/FP64 Performance Ratio: 32
id 1          Tesla T4          [SUPPORTED]
      Compute Capability: 7.5
      PCI Device ID: 5
      PCI Bus ID: 0
      UUID: GPU-5381de6e-5d83-d77f-31c9-4928539835cf
      Watchdog: Disabled
      FP32/FP64 Performance Ratio: 32
Summary:
      2/2 devices are supported
Device ID: 0
Device Name: Tesla T4
Compute Capability: 7.5
Number of Processor: 40
Device's cores: 2560
Memory Size: 14911.69 MB
```

Figure 1: GPU information obtained using `numba.cuda`.